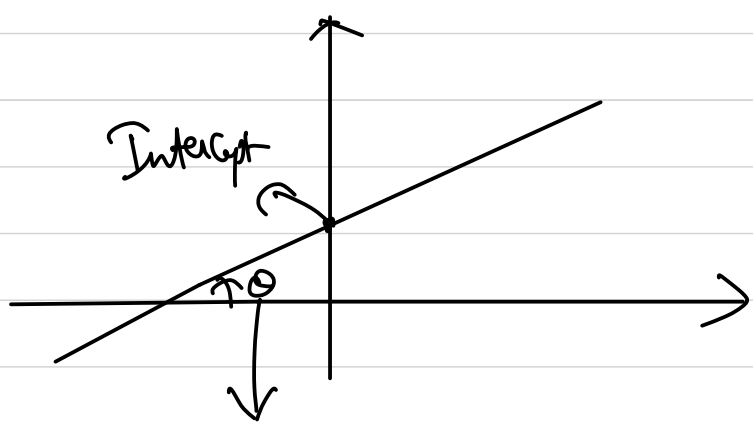
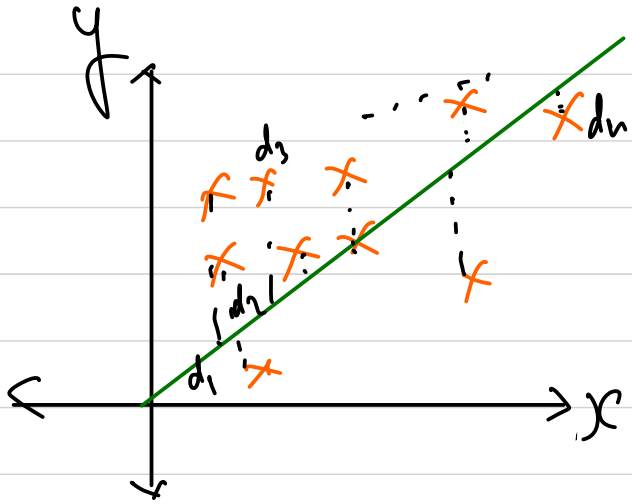
LINEAR REGRESSION

x	y
x_1	y_1
x_2	y_2
x_3	y_3
$\vdots$	$\vdots$
x_n	y_n

$$y = \underset{\text{Slope.}}{m}x + \underset{\text{Intercept}}{b}$$



$m = \tan(\theta)$



$$\text{Error} = \sum_{i=1}^{n} d_i = d_1 + d_2 + d_3 \cdots + d_n$$

We need to find values of m & b

How?

1) Closed form : Set of rigid equations.

2) open form : optimization techniques.

OLS : optimal Least Squared.

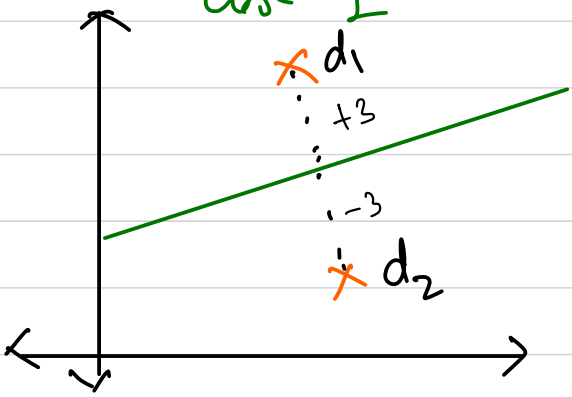
$$\Rightarrow b = \bar{Y} - m\bar{x}$$

$$\Rightarrow m = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

$\bar{x} \rightarrow$ Mean of X

$\bar{Y} \rightarrow$ Mean of y.

Case 1

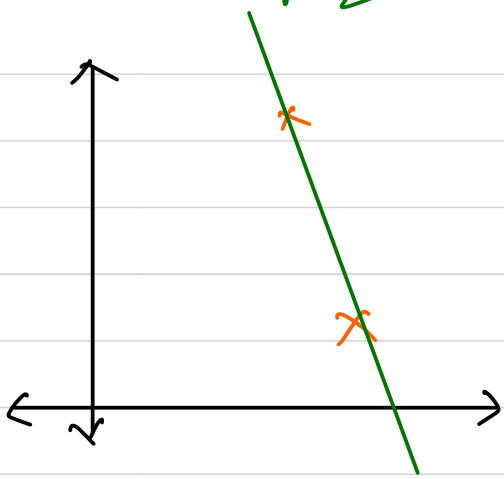


$$\text{Error} = \sum_{i=1}^{n} d_i$$
$$= d_1 + d_2$$
$$= 3 + (-3)$$
$$= 0$$

$$\text{New_Error} = \sum_{i=1}^{n} |d_i|$$
$$= |d_1| + |d_2|$$
$$= |+3| + |-3|$$
$$= 3 + 3$$
$$= 6$$

$$\text{New_Error_2} = \sum_{i=1}^{n} d_i^2$$
$$= d_1^2 + d_2^2 = 3^2 + (-3)^2$$
$$= 9 + 9$$
$$= 18.$$

Case 2



$$\text{Error} = \sum_{i=1}^{n} d_i$$
$$= d_1 + d_2$$
$$= 0 + 0$$
$$= 0.$$

$$\text{New_Error} = \sum_{i=1}^{n} |d_i|$$
$$= |d_1| + |d_2|$$
$$= 0 + 0$$
$$= 0$$

$$\text{New_Error_2} = \sum_{i=1}^{n} d_i$$
$$= d_1^2 + d_2^2 = 0^2 + 0^2$$
$$= 0$$

$$E = \sum_{i=1}^{n} d_i^2$$

$$E = \sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2 \longrightarrow (1)$$

$y \rightarrow$ Ground truth.

$$\hat{y}_i = mx_i + b \longrightarrow (a)$$

$\hat{y} \rightarrow$ Predicted value.

Sub (a) in (1)

$$E = \sum_{i=1}^{n} \left(y_i - (mx_i + b)\right)^2$$

$$E(m, b) = \sum_{i=1}^{n} \left(y_i - (mx_i + b)\right)^2$$

$$E(0, b) = \sum_{i=1}^{n} \left(y_i - b\right)^2$$

$$E(m, 0) = \sum_{i=1}^{n} \left(y_i - mx_i\right)^2$$

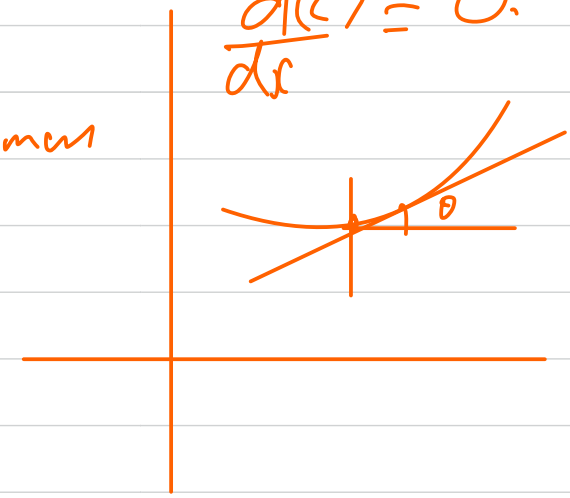
$$y = 3x^2 + 5$$

$$\left(\frac{dx^n}{dx} = nx^{n-1} \right)$$

$$\frac{d(c)}{dx} = 0.$$

Task: Make y - minimum

$y = 5$ iff $x = 0$.



$$\frac{dy}{dx} = \frac{d3x^2 + 5}{dx}.$$

$$= \frac{d\,3x^2}{dx} + \frac{d(5)}{dx}$$

$$\frac{dy}{dx} = 6x + 0$$

For minimum value equal slope to zero.

$$6x + 0 = 0$$

$$x = \frac{0}{6}$$

$$\underline{\underline{x = 0}}$$

$$\frac{dx^n}{dx} = nx^{n-1}$$

$$\frac{dc.x}{dx} = c.$$

$$\frac{d(g(x))}{dx} = \frac{d\,f(x)}{dx} \cdot \frac{d\,g(x)}{dx}.$$

$$E = \sum_{i=1}^{n} (y_i - mx_i - b)^2$$

$$\frac{\partial E}{\partial b} = \sum_{i=1}^{n} \frac{\partial (y_i - mx_i - b)^2}{\partial b}$$

$$= \sum_{i=1}^{n} 2(y_i - mx_i - b)\left(\frac{\partial (y_i - mx_i - b)}{\partial b}\right)$$

$$= \sum_{i=1}^{n} 2(y_i - mx_i - b)\left(\frac{\partial y_i}{\partial b} - \frac{\partial mx_i}{\partial b} - \frac{\partial b}{\partial b}\right)$$

$$= \sum_{i=1}^{n} 2(y_i - mx_i - b)\left(0 - 0 - 1 * b^{(1-1)}\right)$$

$$= \sum_{i=1}^{n} 2(y_i - mx_i - b)(0 - 0 - 1)$$

$$= 2\sum_{i=1}^{n} (y_i - mx_i - b)(-1)$$

$$= -2\sum_{i=1}^{n} (y_i - mx_i - b)$$

To get Minimum Value, we need to equate above to zero.

$$-2\sum_{i=1}^{n}(y_i - mx_i - b) = 0$$

$$\div -2$$

$$\sum_{i=1}^{n}(y_i - mx_i - b) = 0$$

$$\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} mx_i - \sum_{i=1}^{n} b = 0$$

$$\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} mx_i = \sum_{i=1}^{n} b$$

$$\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} mx_i = \underbrace{b+b+b+\dots+b}_{n\text{-times}}$$

$$\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} mx_i = nb$$

$$nb = \sum_{i=1}^{n} Y_i - m \sum_{i=1}^{n} x_i$$

$$b = \frac{\sum_{i=1}^{n} Y_i}{n} - \frac{m \sum_{i=1}^{n} x_i}{n}$$

$$b = \frac{Y_1 + Y_2 + \dots + Y_n}{n} - m \frac{x_1 + x_2 + x_3 \dots x_n}{n}$$

$$\boxed{b = \bar{Y} - m\bar{x}}$$

$$E = \sum_{i=1}^{m} (Y_i - mx_i - b)^2$$

$$E = \sum_{i=1}^{m} \left(Y_i - mx_i - (\bar{Y} - m\bar{x})\right)^2$$

$$E = \sum_{i=1}^{n} (Y_i - mx_i - \bar{Y} + m\bar{x})^2$$

$$\frac{\partial E}{\partial m} = \frac{\partial}{\partial m} \sum_{i=1}^{n} (Y_i - mx_i - \bar{Y} + m\bar{x})^2$$

$$= 2 \sum_{i=1}^{n} \left((Y_i - mx_i - \bar{Y} + m\bar{x}) \left(\frac{\partial (Y_i - mx_i - \bar{Y} + m\bar{x})}{\partial m} \right) \right)$$

$$= 2\sum_{i=1}^{n}\left((y_i - mx_i - \bar{y} + m\bar{x})\left(\frac{\partial y_i}{\partial m} - \frac{\partial mx_i}{\partial m} - \frac{\partial \bar{y}}{\partial m} + \frac{\partial m\bar{x}}{\partial m}\right)\right)$$

$$= 2\sum_{i=1}^{n}\left((y_i - mx_i - \bar{y} + m\bar{x})(0 - x_i\, 0 + \bar{x})\right)$$

$$\frac{\partial E}{\partial m} = -2\sum_{i=1}^{n}\left((y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x})\right)$$

To get minimum Equate above to zero.

$$-2\sum_{i=1}^{n}\left((y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x})\right) = 0$$

$$\sum_{i=1}^{n}\left((y_i - \bar{y} + m\bar{x} - mx_i)(x_i - \bar{x})\right) \overset{\div -2.}{=} 0$$

$$\sum_{i=1}^{n}\left((y_i - \bar{y}) + m(\bar{x} - x_i)(x_i - \bar{x})\right) = 0.$$

$$\sum_{i=1}^{n}\left((y_i - \bar{y}) - m(x_i - \bar{x})(x_i - \bar{x})\right) = 0.$$

$$\sum_{i=1}^{n}\left((y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})(x_i - \bar{x})\right) = 0$$

$$\sum_{i=1}^{n}(y_i-\bar{y})(x_i-\bar{x}) - m\sum_{i=1}^{n}(x_i-\bar{x})^2 = 0$$

$$\sum_{i=1}^{n}(y_i-\bar{y})(x_i-\bar{x}) = m\sum_{i=1}^{n}(x_i-\bar{x})^2$$

$$\boxed{m = \frac{\sum_{i=1}^{n}(y_i-\bar{y})(x_i-\bar{x})}{\sum(x_i-\bar{x})^2}}$$

Home work:

x	y
5	12
6	14
7	16
8	18
9	20
10	22
12	
14	

$m = ?$

$b = ?$

$y = mx + b$

Multiple Linear Regression

X_1 CGPA	X_2 IQ	X_3 Attendance	Y Placement_Package.
3.3 x_{11}	120 x_{12}	80% x_{13}	100K y_1
3.8 x_{21}	110 x_{22}	80% x_{23}	120K y_2
4.0 x_{31}	130 x_{32}	50% x_{33}	160K y_3
3.2 x_{41}	150 x_{42}	90% x_{43}	200K y_4
3.0 x_{51}	100 x_{52}	95% x_{53}	90K y_5
⋮	⋮	⋮	⋮

$$Y = mx + c$$

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

$$\hat{Y}_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_3 x_{13}$$

$$\hat{Y}_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \beta_3 x_{23}$$

$$\vdots$$

$$\hat{Y}_5 = \beta_0 + \beta_1 x_{51} + \beta_2 x_{52} + \beta_3 x_{53}$$

$$\begin{bmatrix} \hat{Y}_1 \\ \hat{Y}_2 \\ \hat{Y}_3 \\ \hat{Y}_4 \\ \hat{Y}_5 \end{bmatrix}_{5\times 1} = \begin{bmatrix} \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_3 x_{13} \\ \\ \\ \\ \beta_0 + \beta_1 x_{51} + \beta_2 x_{52} + \beta_3 x_{53} \end{bmatrix}$$

$$\hat{Y}_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \cdots + \beta_m x_{1m}$$

$$\vdots$$

$$\hat{Y}_n = \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \cdots + \beta_m x_{nm}$$

$$\begin{bmatrix} \\ \hat{Y} \\ \\ \end{bmatrix}_{(n\times 1)} = \overset{X}{\begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} & \cdots & x_{1m} \\ \vdots & & & & & \\ 1 & x_{n1} & x_{n2} & x_{n3} & \cdots & x_{nm} \end{bmatrix}} \overset{\beta}{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}}$$

$$\hat{Y} = X\beta$$

$$\text{Error} = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2$$

$$e = Y - \hat{Y} = \begin{bmatrix} Y_1 - \hat{Y}_1 \\ Y_2 - \hat{Y}_2 \\ \vdots \\ Y_n - \hat{Y}_n \end{bmatrix}$$

$$e^T e = [y_1-\hat{y}_1 \;\; y_2-\hat{y}_2 \; \dots \; y_n-\hat{y}_n] \begin{bmatrix} y_1-\hat{y}_1 \\ y_2-\hat{y}_2 \\ \vdots \\ \vdots \\ y_n-\hat{y}_n \end{bmatrix}$$

Minimize the Error $= e^T e$.

$$= (y-\hat{y})^T (y-\hat{y})$$

$$= (y^T - \hat{y}^T)(y-\hat{y})$$

$$= (y^T y) - (y^T \hat{y}) - (\hat{y}^T y) + (\hat{y}^T \hat{y})$$

$$= y^T y - 2y^T \hat{y} + \hat{y}^T \hat{y}$$

$$\text{Error} = y^T y - 2y^T (X\beta) + (X\beta)^T (X\beta)$$

$$= y^T y - 2y^T X\beta + (X^T \beta^T)(X\beta).$$

To find minima we need to differentiate & equate to zero.

$$\frac{dE}{d\beta} = 0 - 2Y^TX + 2\beta^TX^TX = 0$$

$$\frac{d\,x^TAx}{dx} = 2x^TA.$$

$$\cancel{2}\beta^TX^TX = \cancel{2}Y^TX$$

$$\beta^TX^TX = Y^TX$$

Multiply with $(X^TX)^{-1}$

$$\beta^T(X^TX)(X^TX)^{-1} = (Y^TX)(X^TX)^{-1}$$

$$\beta^T = Y^TX(X^TX)^{-1}$$

$$(\beta^T)^T = \left(Y^TX(X^TX)^{-1}\right)^T$$

$$\beta = \left[(X^TX)^{-1}\right]^TX^TY$$

$$\boxed{\beta = (X^TX)^{-1}X^TY}$$

$$(A^{-1})^T = A^{-1}$$

iff A is symmetric

$$(AB)^T = B^TA$$

Problems with OLS:

→ Very time Consuming.

Optimization

Function ⇒ Mathematical Rule when given an input processes it & gives output.

⇒ Deterministic output → Consistent o/p for a given i/p.

Multivariable Function ⇒ $Z = f(x, y)$ → Depends on multiple inputs.

→ ML Models Can be Considered as Mathematical functions.

* Parametric Vs Non-Parametric Functions:

Parametric	Non-Parametric
Data Distribution Assumption	No Assumption.
Fixed no of Parameters irrespective of # of Rows.	No fixed no of Parameters.
Ex:- Linear/Logistic Regression.	Ex:- Decision Tree.

* Linear Regression as a Parametric function:

CGPA IQ Placement.

Loss Function :

A loss function, also known as a cost function or objective function, is a mathematical function that measures the difference between the predicted output and the actual target values in a machine learning model. The primary goal of training a machine learning model is to minimize the value of the loss function, which corresponds to improving the model's performance on the given task.

Loss functions play a crucial role in the optimization process, guiding the learning algorithm to adjust the model's parameters to achieve better predictions.

→ Minimizing Loss function is the goal of optimization.

* Choosing a Loss Function:

→ Problem type

→ Robustness of outliers.

→ Interpretability & ease of use.

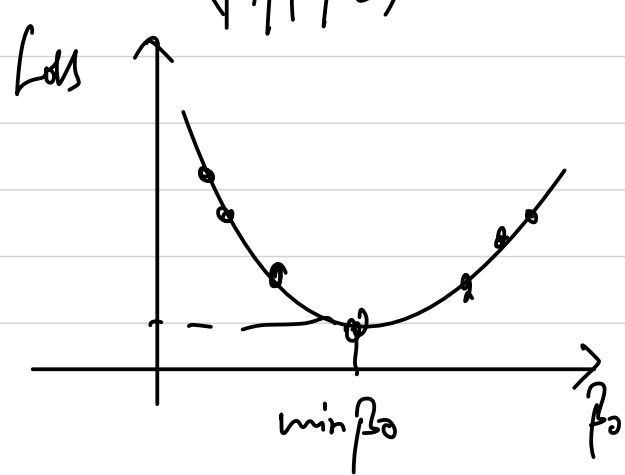
→ Differentiability.

→ Compatibility with the model.

* Calculating parameters from a loss function:

CGPA	IQ	Placement.
8	70	75
7	90	80
6	60	50

$$L(\beta_0, \beta_1, \beta_2) = \underset{(\beta_0, \beta_1, \beta_2)}{\arg\min} \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$



Mathematically to reach minimum point we need to differentiate & equate it to zero.

i.e $\frac{\partial L}{\partial \beta_0} = 0$

$$L = \frac{1}{n}\sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$

$$L = \frac{1}{n}\sum_{i=1}^{n} \left(Y_i - (\beta_0 + \beta_1 X_1 + \beta_2 X_2)\right)^2$$

$$= \frac{1}{n}\sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_1 - \beta_2 X_2)^2$$

$$\frac{\partial L}{\partial \beta_0} = \frac{1}{n}\sum_{i=1}^{n} 2(Y_i - \beta_0 - \beta_1 X_1 - \beta_2 X_2)(-1) = 0$$

$$\frac{1}{n}\sum_{i=1}^{n} Y_i - \beta_1 X_1 - \beta_2 X_2 = \frac{1}{n}\sum_{i=1}^{n} \beta_0$$

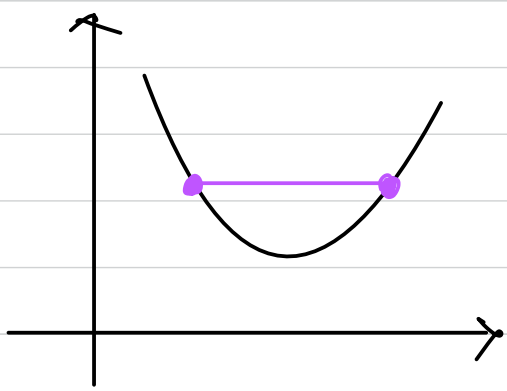
$$\frac{1}{n}\sum_{i=1}^{n} Y_i - \beta_1 X_1 - \beta_2 X_2 = \beta_0$$

$\Rightarrow$ Problem is the above approach leads to local minimas & hence the solution might be incorrect in most of the cases.

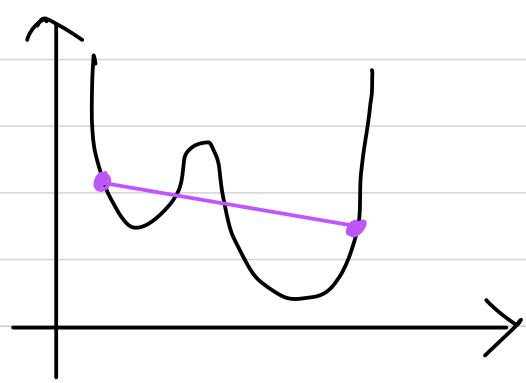
* Problems with above Solution:

1) Non-Convexity,
2) Complexity
3) Scalability
4) Online learning & Streaming data.

* Convex & Non-Convex function :-



Convex
→ Only one Minima



Non-Convex.
→ Multiple local minimas.

Gradient Descent :

To solve above issues we use optimization techniques.

Few optimization techniques.

- → Gradient Descent.
- → Newton Method.
- → Quadratic equations.
- → linear methods.

→ We primarily use Gradient Descent in M.L

→ For SVM & PCA we use Quadratic Programming.

→ $\beta_{0_{new}} = \beta_{0_{old}} - \frac{dL}{d\beta_0}$